

Probabilistic Reasoning over time :
→ already predefined man.
"check yt"

Temporal Model :-

It represents the how a system evolves over time.
Eg: HMM

Inference in Temporal model :

→ means asking question to the model

Types of Inference

→ in temporal model.

"Our model kita shana man
ques kelaan".

Types are,

→ Filtering.

→ Prediction.

→ Smoothing.

→ Most likely sequence.

Image Processing → Filtering plays
major role.

i) Filtering → It is used to find the Probab^{le}
current state given

Eg Prob →

Problem :

yesterday
Rain
No Rain

↓
the

considers,

Emissi

me
R
N

Problem :

yesterday	Today Rain	Today: No Rain
Rain	0.7	0.3
No Rain	0.3	0.7

↓
this is transition Probability.

consider, $t \rightarrow$ today

$t-1 \rightarrow$ yesterday

$t+1 \rightarrow$ tomorrow

So consider, given

$$P(\text{Rain}_t | \text{Rain}_{t-1}) = 0.7$$

$$P(\text{No Rain}_t | \text{Rain}_{t-1}) = 0.3$$

Emission Probability: It is also known as

unseen model.

weather	umbrella	No umbrella
Rain	0.9	0.1
No Rain	0.2	0.8

$$P(\text{umbrella} | \text{Rain}) = 0.9$$

$$P(\text{No umbrella} | \text{Rain}) = 0.1$$

$$P(\text{umbrella} | \text{No Rain}) = 0.2$$

$$P(\text{umbrella} | \text{no Rain}) = 0.8 //$$

Initial Probability :

$$P(\text{Rain}) = 0.5$$

$$P(\text{No Rain}) = 0.5$$

Given that,

the person carry the umbrella today what is probability that raining today?

$$P(\text{Rain} | \text{umbrella}) = \frac{\text{Prior} \times \text{likelihood}}{\text{Evidence}}$$

no divide (keep in ur mind)

using Baye's theorem,

$$P(\text{Rain} | \text{umbrella}) = P(\text{umbrella} | \text{Rain}) \times P(\text{Rain})$$

$$\begin{array}{r} 1.0 \\ 0.45 \\ \hline 1.55 \end{array}$$

$$\begin{array}{r} \text{not raining} = 1 - 0.45 \\ \hline = 0.55 \end{array}$$

$$P(\text{No Rain} | \text{umbrella}) = P(\text{umbrella} | \text{No Rain})$$

$$= 0.2 \times 0.5$$

$$= 0.10$$

So the both values we have to normalize it

$$\text{Normalization} = \frac{0.45}{0.45 + 0.10} = \frac{0.45}{0.55}$$

$$= 0.8182$$

$$(\text{No Rain} | \text{umbrella}) = \frac{0.10}{0.45 + 0.10} = \frac{0.10}{0.55}$$

$$= 0.1818$$

So if we add both values, it gives "1"

So its value is

ii) Prediction:

It used to find the Probability of future state of given current evidence.

Eg:- a Person carrying an umbrella given today's ~~observation~~, what is the Probab that it Rain tomorrow "will"

When,

$PCRain_{t+1} | \text{umbrella}$

$$\Rightarrow P(Rain_{t+1} | Rain_t) * \overset{0.7}{P(Rain_t)}$$

$$\overset{0.8182}{P(Rain_t)},$$

$$+ P(Rain_{t+1} | \text{NO Rain}_t) * \overset{0.3}{P(\text{NO Rain}_t)}$$

$$\overset{0.1818}{P(\text{NO Rain}_t)}$$

$$\Rightarrow 0.7 * 0.8182 + 0.3 * 0.1818$$

$$\Rightarrow 0.62728 //$$

Smoothing:

This is used to find the Probability of Past state given observation

State (Past)
 $P(X_k | E_{1:t})$

all the evidence

$$k < t$$

upto today

It will

be lesser than 1

It uses two Probability

→ forward Probability

→ Backward Probability.

Ques:

given that the Person did not carry an umbrella on day one and carried the umbrella on day 2. What is Probab that rainy on day 2.

Step 1:

finding the forward Prob

$P(X_1 | E_1)$ (Previous day state)
using observation

$$P(x_1 | e_1) \rightarrow \text{Day 1}$$

"Day 1 no umbrella"

$e_1 = \text{no umb}$

$$P(x_1 | e_1) \rightarrow P(\text{Rain}_1 | \text{no umb}_1)$$

$$\Rightarrow P(\text{no umb} | \text{Rain}_1) * P(\text{Rain}_1)$$

$$= 0.1 * 0.5$$

$$= 0.05 //$$

$$P(\text{no Rain}_1 | \text{no umb}_1) = P(\text{no umb}_1 | \text{no Rain}_1) * P(\text{no Rain}_1)$$

$$= 0.8 * 0.5$$

$$= 0.4 //$$

$$= 0.40 //$$

So we have to normalize,

$$= \frac{0.05}{0.05 + 0.40}$$

$$= \frac{0.05}{0.45}$$

$$= 0.1111$$

$$\Rightarrow \frac{0.40}{0.05 + 0.40} = \frac{0.40}{0.45}$$

$$= 0.8889 //$$

$$\text{Plus } \left(\begin{array}{r} 0.1111 \\ 0.8889 \end{array} \right)$$

$$\times \left(\begin{array}{r} 1.0000 \\ 1.0000 \end{array} \right) = (1.0000)$$

Step 2:

calculating Backward Probability

$$P(E_2 | X_1)$$

Evidence

Day 2

Rain or no
Rain
Pay 1

assume,

$$P(E_2 | X_1) \rightarrow$$

Consider as Rain on
day 1

$$P(E_2 | X_1) \rightarrow P(Rain_2 | Rain_1) * P(umbrella_2 | Rain_2) +$$

$$P(NoRain_2 | Rain_1) * P(umbrella_2 | NoRain_2)$$

$$P(NoRain_2 | Rain_1) * P(umbrella_2 | NoRain_2)$$

$$(umbrella_2 | NoRain_2)$$

Prob
rain

prob
not
rain

$$\Rightarrow (0.7 \times 0.9) + (0.3 \times 0.2)$$

$$(0.63) + (0.06)$$

$$= 0.69$$

↳ Prob carrying umbrella
in Rain

$$P(E_2 | X_1) = P(\text{Rain}_2 | \text{No Rain}_1) \times$$

$$P(\text{umb}_2 | \text{Rain}_2) +$$

$$P(\text{no rain}_2 | \text{No Rain}_1) \times P(\text{umb}_2 | \text{No Rain}_2)$$

$$= 0.3 \times 0.9 + 0.7 \times 0.2$$

$$= 0.27 + 0.14$$

$$= 0.41$$

↳ Probab carrying umb in
no Rain.

Step 3:

combining both forward and backward.

$$P(X_1 | E_1, E_2) \propto P(X_1 | E_1) \times P(E_2 | X_1)$$

For Rain =

For Not

Normalizati

Rain

no Ra

PD) Most Likely

o Elevation

$$\begin{aligned}\text{For Rain} &= \cancel{0.05} \times \cancel{0.69} \\ &= 0.1111 \times 0.69 \\ &= 0.07666\end{aligned}$$

$$\begin{aligned}\text{For Not Rain} &= 0.8889 \times 0.41 \\ &= 0.36445 \text{ //}\end{aligned}$$

Normalization:

$$\begin{aligned}\text{Rain} &= \frac{0.07666}{0.36445 + 0.07666} & \text{Approx} \\ &= 0.1738\end{aligned}$$

$$\begin{aligned}\text{no Rain} &= \frac{0.36445}{0.07666 + 0.36445} \\ &= 0.8262 \text{ Approx.}\end{aligned}$$

Q) Most likely sequence :- "Viterbi Algorithm"
It is used to calculate the sequence